CS/EE-712: Image Processing

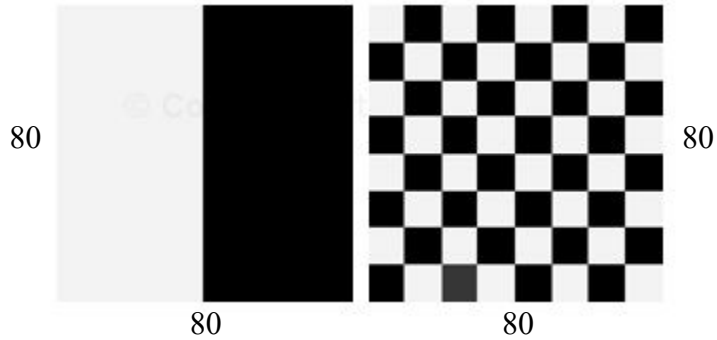
Homework

Assigned: March 13, 2026 (Friday)
Due on Canvas: 11:59PM on March 27, 2026 (Friday)

Problem 1 (5 points) Textbook **Problem 4.7**

As the figure below shows, the Fourier transform of a "tent" function (on the left) is a squared sinc function (on the right). Advance an argument that shows that the Fourier transform of a tent function can be obtained from the Fourier transform of a box function. (*Hint:* The tent itself can be generated by convolving two equal boxes.)

Problem 2 (5 points)
The images shown below are quite different, but their histograms are the same. Suppose that each image is blurred with a 3*3 averaging mask. Would the histograms of the blurred images still be equal? Please explain.



Problem 3 (5 points)

Consider the image segment shown.

(a) Let $V = \{0, 1\}$ and compute the lengths of the shortest 4- and 8-path between p and q. If a particular path does not exist between these two points, explain why.

(b) Repeat for $V = \{1, 2\}$.

$$\begin{array}{cccc} 3 & 1 & 2 & 1\,(q) \\ 2 & 2 & 3 & 2 \\ 1 & 2 & 1 & 1 \\ (p)\,1 & 0 & 1 & 2 \end{array}$$

Please note that this image is slightly different from problem P2.15 in the textbook.

Problem 4 (5 points)

Perform histogram equalization on the following histogram, where r is the intensity level and n is the number of pixels with the corresponding intensity level. You need to find out the histogram after the equalization.

r	0	1	2	3	4	5	6	7
n	400	700	800	900	500	400	196	200

Problem 5 (5 points) Textbook **Problem 3.27**

Give a 3*3 mask for performing unsharp masking in a single pass through an image. Assume that the average image is obtained using the filter below:

$$\frac{1}{9} \times \begin{array}{|c|c|c|} \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline \end{array}$$

Problem 6 (5 points) Textbook **Problem 3.28**
Show that subtracting the Laplacian from an image is proportional to unsharp masking. Use the definition for the Laplacian given below:

$$\nabla^2 f(x, y) = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$

Problem 7 (5 points): Textbook **Problem 4.12**
Consider a checkboard image in which each square is 1mm * 1 mm. Assuming that the image extends infinitely in both coordinate directions, what is the minimum sampling rate (in samples/mm) required to avoid aliasing?
(Hint; consider the image in the form of . . . 101010101 . . . .The period of this signal is 2mm.)

Problem 8 (5 points): Textbook **Problem 4.27**
Consider a 3*3 spatial mask that averages the four closest neighbors of a point (x, y), but excludes the point itself from the average.

(a) Find the equivalent filter, H(u, v), in the frequency domain.
(b) Show that your result is a low-pass filter.